

# **The Effects of Brownfield Designation on Parcel Tax Valuation in Worcester, MA**

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# Agenda

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# Introduction

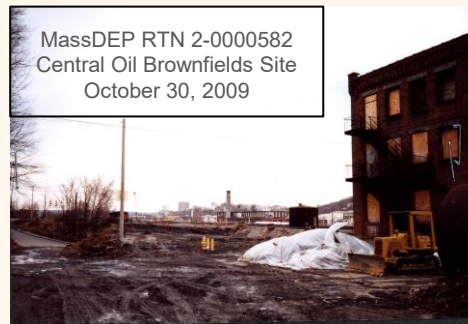
## What are “Brownfields”?

- EPA definition of Brownfields: a property for which the expansion, redevelopment, or reuse ... may be complicated by the presence or potential presence of a hazardous substance, pollutant, or contaminant ([Linn, 2012](#))
- Cincinnati, OH case study, 0.1% ↓ property value with each percent of property that was located near the BF ([Turečková et al., 2022](#))
- Meta-analysis of US based studies found that the average residential property within 2 miles from a source of environmental contamination loses 9.5% of its value – property values losses were also \$0.23 higher for every additional dollar in home value ([De Sousa et al., 2009](#))
- BF has overall negative impact on property market, demand for properties drop ([Mihaescu et al., 2012](#))

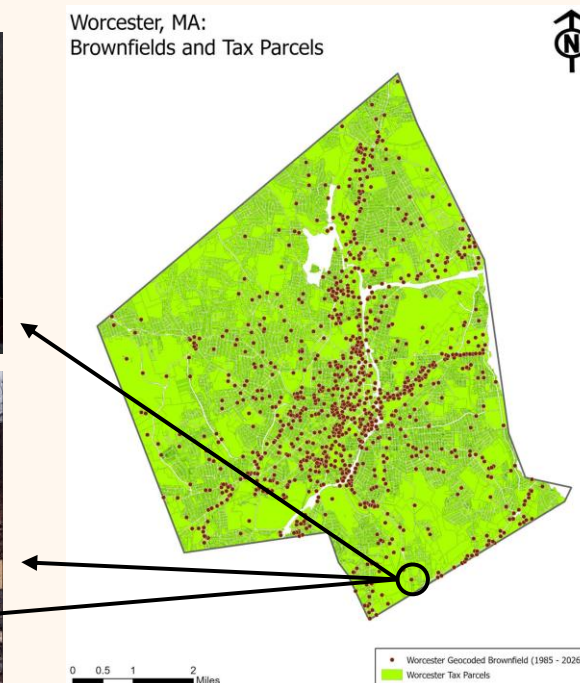
# Introduction

## Project Area —

- 47,409 parcels in 2025
- 1,715 BF's from 1985 - 2026



Worcester, MA:  
Brownfields and Tax Parcels





# Broad Research Scope

## Do brownfields affect parcel tax valuation in Worcester, MA?

### Specific Questions

- Do brownfields affect parcel tax valuation? If so, what is the relationship between predictors and the response variable based on a multi-linear regression model?
- What do model diagnostics indicate regarding linearity of these relationships?
- Is tax valuation significantly spatially autocorrelated?



# Broad Research Scope

## Variables

### Response

- Tax valuation (dollars)

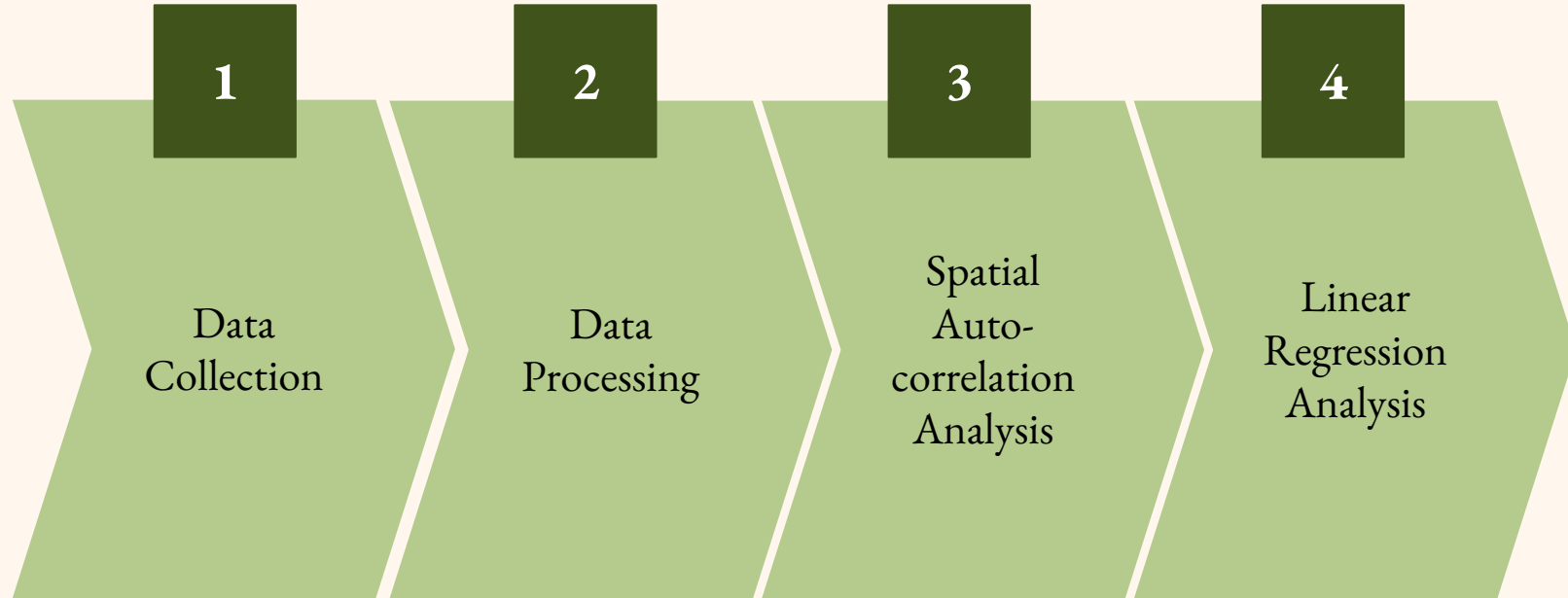
### Predictors

- Recent list price (dollars)
- Lot size (acres)
- Proximity to brownfields (500ft)
- Duration of designation (days)
- Duration of certification (inactive)
- Contaminant (oil / oil and hazmat)

# Dataset Introduction

| <u>Dataset</u>                     | <u>Source</u>                                                                            | <u>Description</u>                                                                          |
|------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Waste Site and Reportable Releases | Massachusetts Executive Office of Energy and Environmental Affairs Data Portal (MassDEP) | This dataset includes information about all of the designated brownfields in Massachusetts. |
| Property Tax Parcels               | MassGIS                                                                                  | This dataset includes tax assessor attributes for land parcels in Massachusetts             |

# Methods





# Methods

## Download data from MassDEP (EEA Portal)

Filtered dataset to Worcester MA and downloaded on 03/16/2026.

## Spatial Join to Worcester Property Tax Parcels

Joined brownfields points to Worcester parcels within 500ft (one to one), created new field to count unique RTN joins

## Export Table

pbf\_regression\_tabel.csv

## Geocoded brownfields dataset using ArcGIS Pro

Removed points outside of Worcester and moved points to nearest parcel if outside polygon (interstate, street, pond, etc)

Created fields: Duration Days for active RTNs (python calc for [03/16/2026] - [Notification date], Inactive Days (python calc for [03/16/2026] - [Notification date]), and Contam Types (Dummy)

## QC Table

Removed top seven RTNs join count rows as obvious outliers (RR parcels that span whole city with hundreds of joins each)

Removed parcels with TOTAL\_VAL = 0 or null

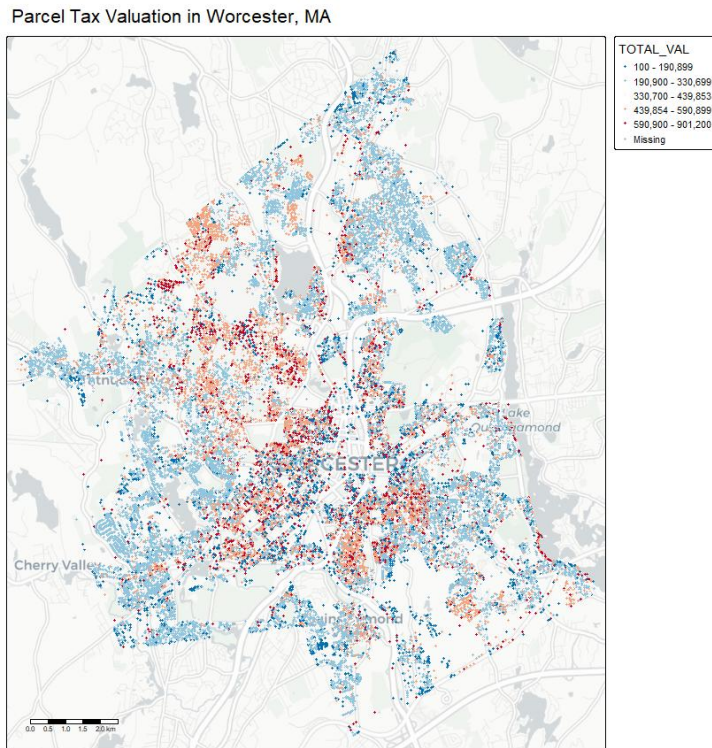
All null values given 0 value for fields: L\_Sale, RTNs, Duration, Inactive

# Spatial Autocorrelation Methods

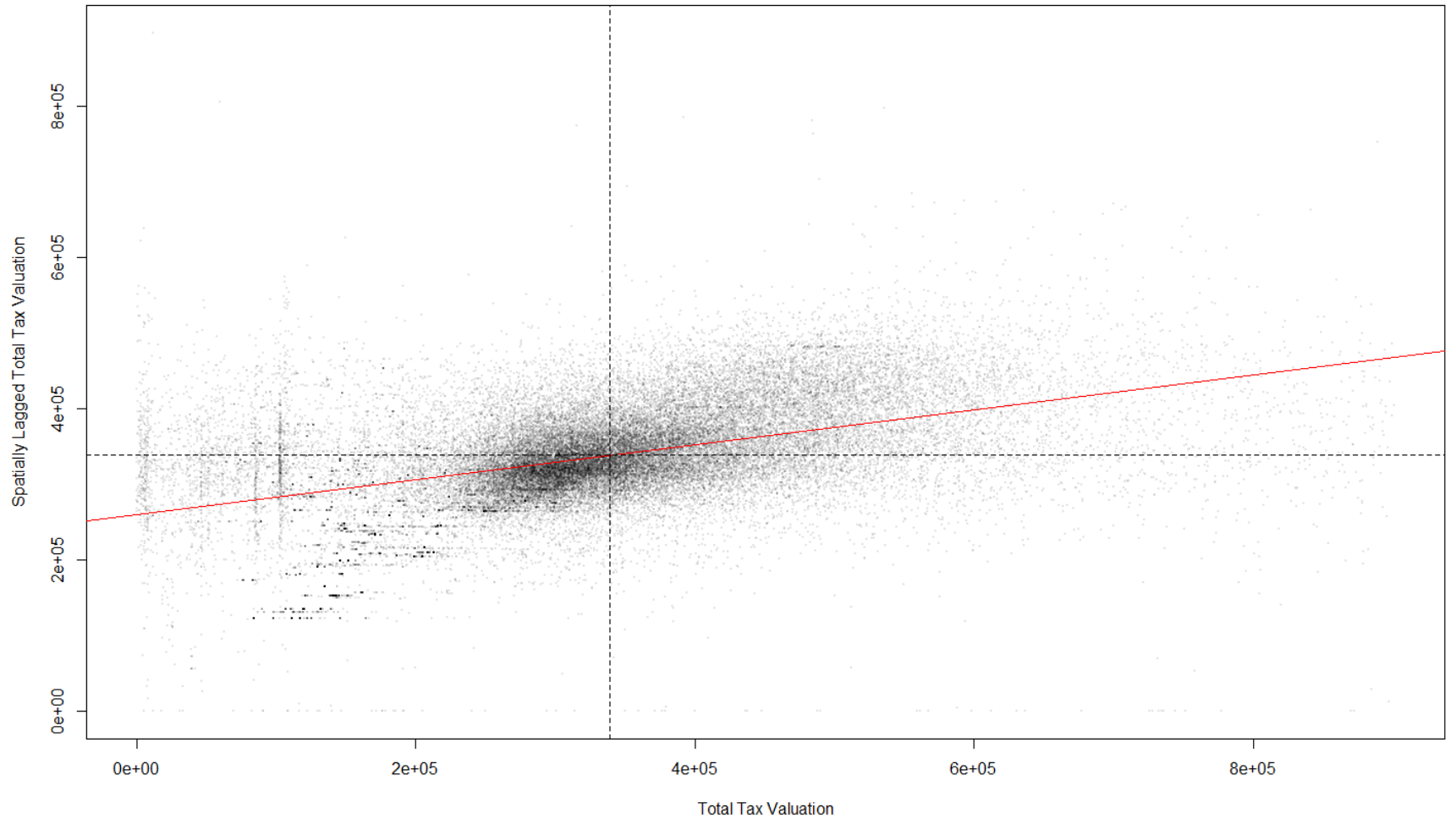
## In-Class Concepts Used

- Remove outliers and values that equal 0 from TOTAL\_VAL:
  - 1) IQR Method:  $IQR = Q3 - Q1$
  - 2) Upper Fence =  $Q3 + (3 \times IQR)$ , values  $>$  upper fence were removed
  - 3) 2,048 of observations removed
- Define spatial relationships, choose neighbor rule, and construct spatial weights matrix
  - Distance = 152 m, or about 500 ft (nb\_dist = 152)
- Compute spatial autocorrelation statistics
  - Global Moran's I value and Moran's scatterplot
  - Local Moran's I values and LISA Cluster map

# Spatial Autocorrelation Figures

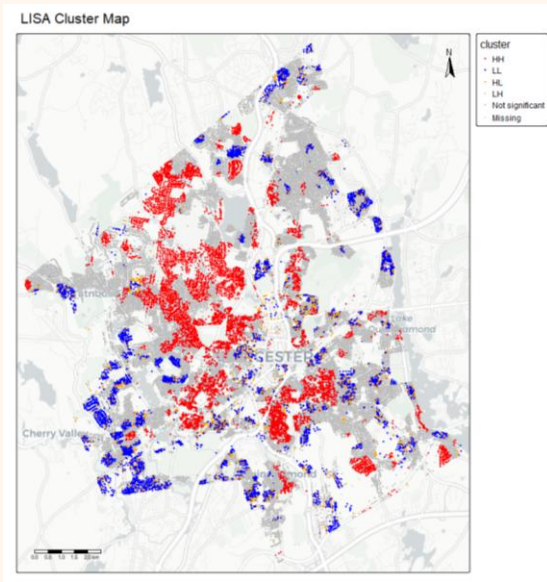


Moran's I Scatterplot

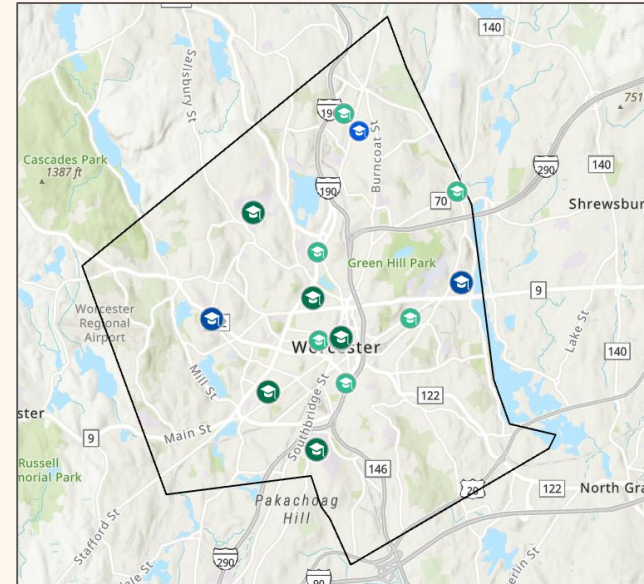




# Spatial Autocorrelation Figures (Cont.)



LISA Cluster Map of Worcester for Parcel Tax Valuation



Map of universities and colleges in Worcester, MA

# Spatial Autocorrelation Disc. of Results

## What does the data mean?

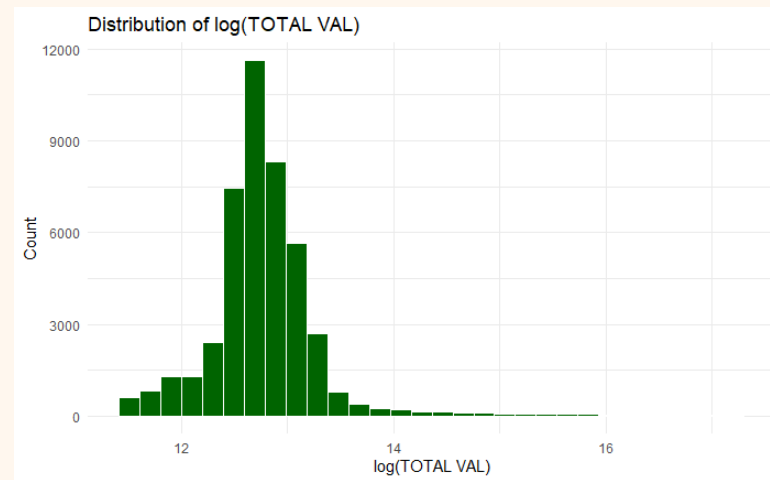
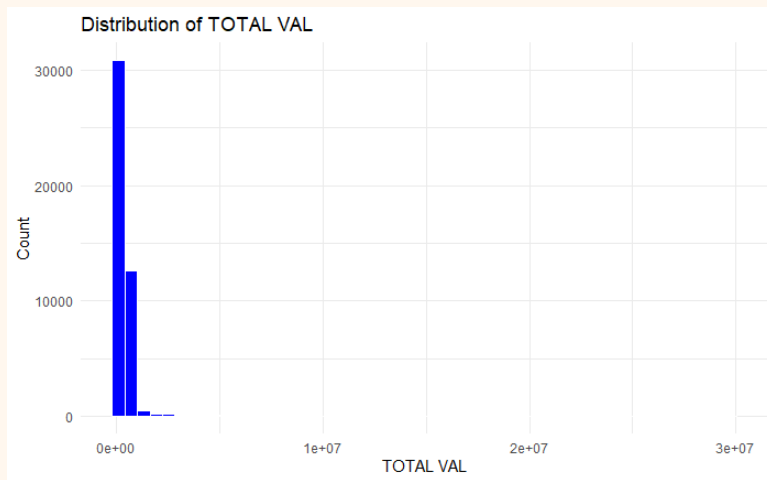
- For the city of Worcester, there is a statistically significant low positive spatial autocorrelation (Global Moran's  $I = 2.311620e-01$ )
- Local Moran's  $I$ 
  - Clustering of HH and LL values throughout the study area
  - High tax valuation of parcels associated with academic institutions (HH values)
  - Surrounding low income neighborhoods (LH and LL values)

# Total Valuation Distribution w/in 95.5%

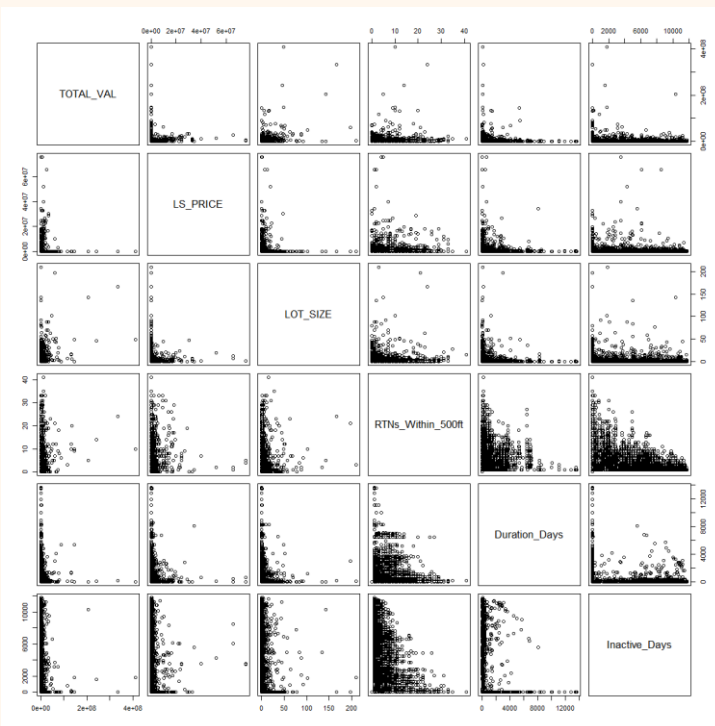
#R-Squared = .1499, keeps 95.5% of data

```
Tot_Val <- Tot_Val %>%
```

```
  filter(TOTAL_VAL >= 100000 & TOTAL_VAL <= 300000000)
```



# Correlation Scatterplot





# Multi-Linear Regression Model Fit (Normal)

Total tax val =  $\beta_0 + \beta_1 \text{List Price} + \beta_2 \text{Lot Size} + \beta_3 \text{BFs within 500 ft} + \beta_4 \text{Duration} + \beta_5 \text{Inactive} + \beta_6 \text{OilYES} + \beta_7 \text{Oil and HazMatYES} + \epsilon$

| Variable                                            | For every one unit increase in each predictor tax valuation increases by this amount: |
|-----------------------------------------------------|---------------------------------------------------------------------------------------|
| Recent listing sale price                           | \$0.2993                                                                              |
| Lot size                                            | \$46,020                                                                              |
| Amount of brownfields within 500 ft                 | \$46,120                                                                              |
| Amount of days the brownfield has been active       | -\$18.23                                                                              |
| Amount of days since the brownfield was closed      | -\$5.13                                                                               |
| Being contaminated with oil                         | \$62,170                                                                              |
| Being contaminated with oil and hazardous materials | \$84,760                                                                              |

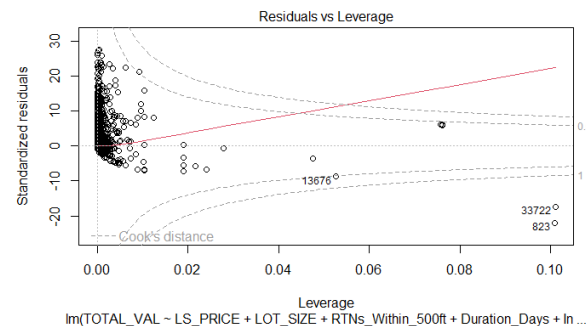
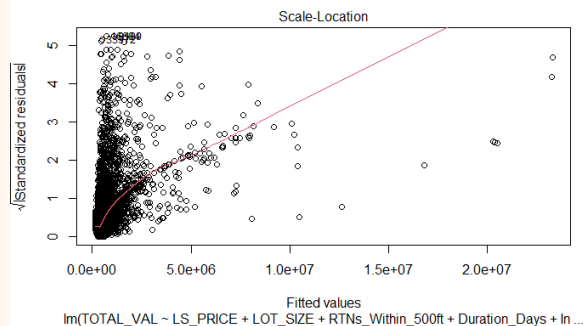
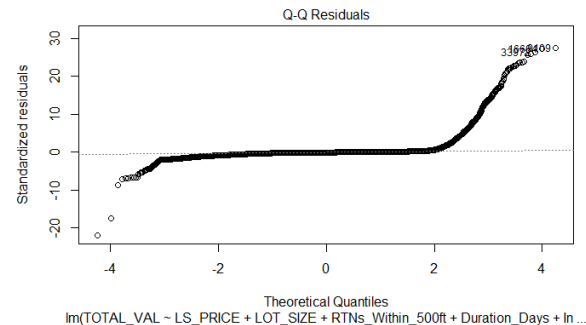
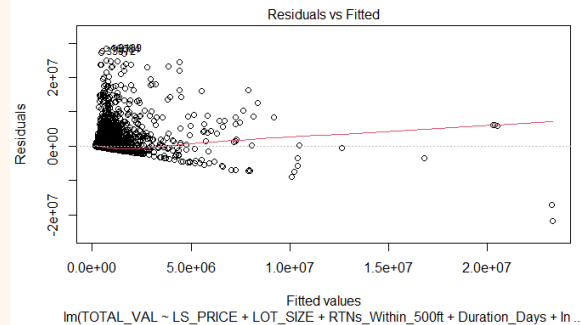
% variance explained by model — 14.99%

# Collinearity Test

| Variable                                                                  | VIF  |
|---------------------------------------------------------------------------|------|
| Recent listing sale price                                                 | 1.01 |
| Lot size                                                                  | 1.00 |
| Amount of brownfields within 500 ft                                       | 1.32 |
| Amount of days the brownfield has been active                             | 1.12 |
| Amount of days since the brownfield was closed (is that what that means?) | 1.76 |
| Being contaminated with oil                                               | 2.23 |
| Being contaminated with oil and hazardous materials                       | 1.28 |

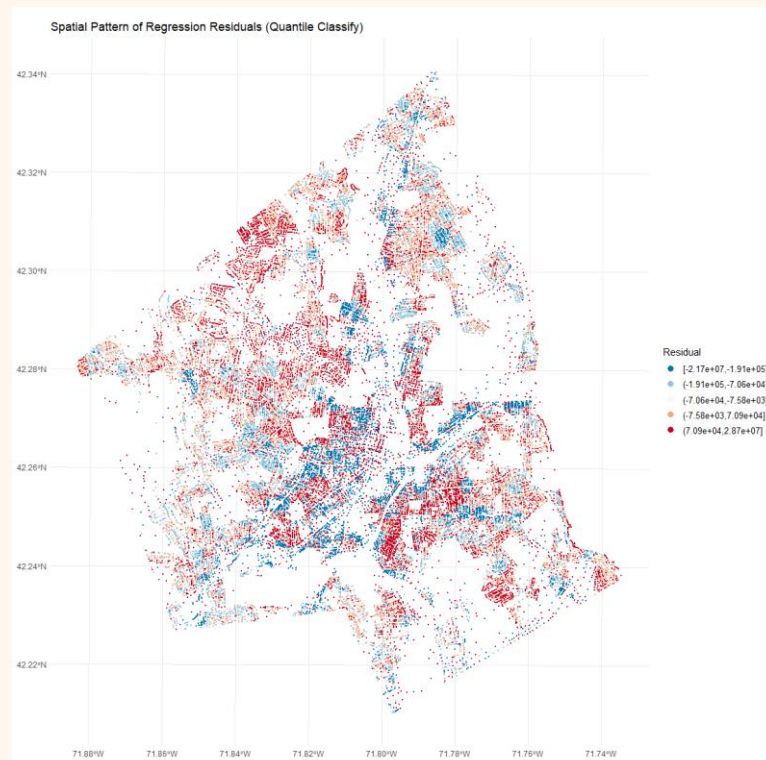
- According to the VIF results above, there is no significant collinearity identified in this model
- No value over 2.228956, which is considerably less than a problematic value of between 5-10 or greater

# Regression Residual Diagnostic Plots



# Spatial Pattern of Regression Residuals

- Spatial correlation of the regression model residuals appears mild
- Clustering of high value residuals around university areas
- Clustering of low residual values near the center of town and along Main South





# Assessment of Linear Regression Results

| Assumption |                              | Conform / Violate | Description                                                                                                                                                                                                                                                                              |
|------------|------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1          | Linearity                    | ✗                 | <ul style="list-style-type: none"><li>Residual vs. fitted plot appears to show weak linearity</li><li>Data is clearly right-skewed</li><li>Linear model is not a good fit</li></ul>                                                                                                      |
| 2          | Independence of Errors       | ✗                 | <ul style="list-style-type: none"><li>Correlation of error terms</li><li>Global Moran's I and LISA Cluster map confirm mild spatial autocorrelation</li><li>Spatial pattern of regression residuals confirms mild clustering</li></ul>                                                   |
| 3          | Constant Variance of Errors  | ✗                 | <ul style="list-style-type: none"><li>Non-constant variance of error terms</li><li>Scale-Location plot confirm that there is a significant level of heteroscedasticity (non-constant variance in residuals) in this dataset</li></ul>                                                    |
| 4          | Normality of Errors          | ✗                 | <ul style="list-style-type: none"><li>Q-Q Residuals plot shows heavy tails - errors are not normally distributed</li><li>Residuals vs. Leverage plot confirms outlier and high-leverage points remain in dataset despite data processing/QC/filtering, though not as prevalent</li></ul> |
| 5          | No Perfect Multicollinearity | ✓                 | <ul style="list-style-type: none"><li>VIF function results show no value over 2.228956</li><li>Scatterplot matrix and matrix of correlation show low correlation between predictors</li><li>No significant collinearity in this dataset (HazMat removed as aliased)</li></ul>            |

# Citations

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1. De Sousa, C., A., Wu, C., Westphal, L., M., 2009. Assessing the Effect of Publicly Assisted Brownfield Redevelopment on Surrounding Property Values. *Economic Development Quarterly*, 23 (2), 95-110.
2. Gillmor, P., E., 2001. *Small Business Liability Relief and Brownfields Revitalization Act* [online]. United States Environmental Protection Agency. Available from: <https://www.congress.gov/bill/107th-congress/house-bill/2869> [Accessed 04/26/2026].
3. Linn, J., 2012. The Effect of Voluntary Brownfields Programs on Nearby Property Values. *Resources for the Future*, Discussion Paper, 12-35.
4. Mihaescu, O., & vom Hofe, R. (2012). The impact of brownfields on residential property values in Cincinnati, Ohio: A spatial hedonic approach. *Journal of Regional Analysis & Policy*, 42(3).
5. Turečková, K., Martinát, S., Nevima, J., & Varadzin, F. (2022). The Impact of Brownfields on Residential Property Values in Post-Industrial Communities: A Study from the Eastern Part of the Czech Republic. *Land*, 11(6), 804. <https://doi.org/10.3390/land11060804>

# Thank you!

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*Questions?*